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Abstract—

I. INTRODUCTION

the detector would still output prediction, but it is not safe in those condition we didnt train for. so it is also important to know when should we stop, apart from having the high classification and localization accuracy from the detector.

we found that, The earlier stage is more sensitive to the corruption in the image than in the later stage, which is more tuned for the detector output.

different corruption have different effect on the channel reaction. For example,... (example a bit why, some are increasing the average reaction, some are making the variance bigger)

At last, we were using the first stages to calcualte the channel behavior, compare them with the closest images in the cleaning image bank (selected with the later stage, which is more about he scenario setting of the image, because it doesnt seem reasonable to compare a beach image with some busy road image for example, they definitely have very different channel reaction, and that's why we choose to use closer images.)

We validate the result for different detectors... (some numbers and trend)

II. RELATED WORK

A detector that is accurate on its test set can still fail on the road, where it meets objects and conditions it was not trained for, and it keeps producing detections all the same. Such failures, which the safety of the intended functionality (SOTIF) addresses [1], call for a runtime monitor that tells the vehicle when to stop trusting its detector [2], [3]. Monitors differ in the question they answer. Out-of-distribution (OOD) monitors ask whether the input contains an object of a class the detector was never trained on [4], [5]. Uncertainty monitors and failure predictors ask how far the detector's output can be trusted [6]–[9]. Covariate-shift monitors [10] ask whether the input itself has changed while its content stays familiar, as when the image is foggy, blurred, noisy, or compressed [11], [12]. Our work belongs to this third kind. A monitor of this kind can take its signal from one of three places: the detector's predictions (Sec. II-A), the image (Sec. II-B), or the detector's features (Sec. II-C).

A. Monitoring the Prediction

Ideally, a model should be less confident on inputs it is not familiar with [13]. The detector's outputs are therefore a natural place to start. SAOD [14] averages the uncertainty of the three most confident detections to decide whether to accept an image. ContrastiveConf [15] subtracts a weighted mean confidence of the queries a DETR [16] discards from the

mean confidence of those it keeps. Beyond the raw confidence, OOD scores computed from a classifier's logits, such as the energy [17] and the maximum logit [18], follow the same idea. These scores react only once the corruption has disturbed the detections, so a mild corruption is easily missed (Sec. IV).

B. Monitoring the Image

The image itself can be judged by a model beside the detector. Image-quality assessment (IQA) models predict how degraded an image looks. Image-based monitors differ in what they learn from: clean images, degraded images, or neither. Models that learn from clean images flag departures from them. NIQE [19] measures how far an image's local contrast statistics differ from those of clean photographs. DisCoPatch [20] trains a discriminator to reject the imperfect patches of its own variational autoencoder, and flags an image whose patches the discriminator rejects. Models that learn from degraded images either recognize particular conditions from labeled examples, such as fog [21] or rain and snow [22], or learn, from synthetically distorted images and human ratings of them, one quality score that applies to any type of degradation, as BRISQUE [23] and ARNIQA [24] do. CLIP-IQA [25] needs no quality-specific training: it rates an image zero-shot with a pretrained CLIP [26], by comparing it with the prompts "Good photo" and "Bad photo" All of these run a second model beside the detector and leave unused the features the detector has already computed.

C. Monitoring the Detector's Features

Finally, a monitor can take its signal from the detector's internal features. Hashemi et al. [27] fit a mean and spread to each neuron on clean images, and flag an image when more neurons than usual leave mean $\pm$ about two standard deviations. Becker et al. [28] compare each channel's activation distribution in a frozen detector with the channel's distribution over the training set. A learned alternative fine-tunes a separate copy of the detector's backbone, with a small embedding head, on synthetically degraded images, and flags an image whose embedding lies far from that of clean images [29].

Like Hashemi et al. [27] and Becker et al. [28], our monitor needs only the frozen detector and clean images. Their single global reference, however, mixes two sources of variation, the scene and the corruption. A mild corruption can thus hide within the differences between clean scenes. Our monitor therefore compares an image's early channels only with those of clean scenes of similar content, found by the detector's deepest stage, which reacts least to corruption yet still describes the scene (Sec. III). It also separates two effects of corruption on these channels: a flattening, where

their strongest responses fall, and a re-weighting, where their levels move up and down.

III. METHOD

The detector is trained on a training set whose images follow a distribution P . At test time, it receives clean test images x , assumed to follow the same distribution P , and their corrupted versions $\tilde{x} = T(x)$, where T is an unknown corruption such as fog, blur, noise or compression. We seek a score $s(\cdot)$ that is higher for corrupted test images $\tilde{x}$ than for clean test images x . The score may draw on only two sources. The first is the set of features that the frozen detector already computes on the test image. Reading only these keeps the validated detector unchanged and adds no second network on every frame. The second is a set $\mathcal{R}$ of clean reference images, taken from the training set, that shows how these features behave on clean images. Thus s is fitted without any corrupted images.

We build s by answering four questions in turn, one per subsection, namely what to measure on a feature map (Sec. III-A), at which stages of the backbone to measure it (Sec. III-B), what to compare the measurements with (Sec. III-C), and how to combine the comparisons into one score (Sec. III-D). The method was developed on RT-DETRv2 [30]. Sec. IV later shows that it applies to other detectors as well.

A. What to Measure

Object detectors typically compute feature maps with a backbone and then turn them into detections with modules specific to their design, such as the region proposal and box heads of two-stage detectors [31], [32] or the transformer decoder of DETR-style detectors [16], [33]. We measure on the backbone for two reasons. First, along a detector, the features shift from describing the image to describing its objects. A corruption changes the image’s appearance, not its objects, so its trace is strongest before the detection-specific modules, in the backbone. Second, these modules differ from design to design, whereas a backbone is common to them, so a monitor built on it carries over across detectors. The backbone computes its maps at stages $\ell = 1, \dots, L$, where a transformer block counts as one stage. We write $a_{c,n}^\ell(x)$ for the activation of channel c at position $n \in \{1, \dots, N_\ell\}$ of stage ℓ .

A channel responds to one visual pattern, such as an edge or a texture [34]. Its value at a single position is not comparable across images, since the same position may show sky in one image and a car in the next. We therefore summarize each channel over all of its positions by two numbers, its *mean response* μ and its *peak-to-mean ratio* ρ , defined as

$$\mu_c^\ell(x) = \frac{1}{N_\ell} \sum_{n=1}^{N_\ell} |a_{c,n}^\ell(x)|, \quad \rho_c^\ell(x) = \log \frac{p_c^\ell(x)}{\mu_c^\ell(x)}, \quad (1)$$

where the *peak response* $p_c^\ell(x)$ is the mean of $|a_{c,n}^\ell(x)|$ over the channel’s strongest 1% of positions, and the absolute value covers signed activations such as transformer tokens. Thus μ captures how strongly a channel responds, and ρ how its response is distributed.

A corruption can change a channel’s responses in two ways. Corruptions that remove contrast or sharpness, such as fog and blur, weaken the edges and textures on which a channel’s strongest responses lie. Their peak responses therefore tend to fall, while their mean responses change much less, so ρ drops. Corruptions that add fine structure, such as noise, tend to move the mean responses instead, in both directions. The first layer’s rectification turns the zero-mean noise into extra response and raises the mean response of nearly every first-layer channel. Later layers combine their inputs with positive and negative weights and turn this into rises in some channels and falls in others. Fig. 1 illustrates both trends on one channel, whose peak response falls by 34% under fog and whose mean response rises by 56% under noise. Across the COCO validation images [35], fog, contrast reduction and the five blur types lower ρ in 75–84% of the stage-1 channels, for mild and strong corruptions alike. The four noise types lower it in only about half of the channels, but move the mean responses of 72–92% of them by more than 10%, up or down. Since corruptions change the two numbers in different ways, we compute both for every channel and next ask where in the backbone to compute them.

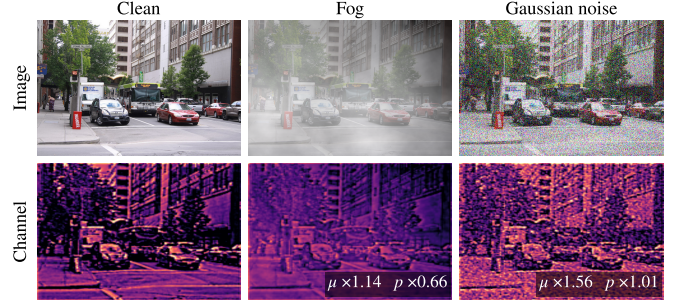


Fig. 1. A backbone channel of RT-DETRv2-R18 on one image. Top: clean image, image under fog and image under Gaussian noise. Bottom: the channel’s response maps, where brighter means a stronger response. Relative to the clean image, fog lowers the peak response ρ , and noise raises the mean response μ .

B. Where to Measure

Within the backbone, the early stages describe how the image looks, and the later stages increasingly describe the objects the detector has to find. A corruption changes how the image looks, so we expect it to show most in the early stages. To find where corruptions show most, we measure how far a corruption moves each channel, relative to how much the channel varies between clean scenes. For channel c of stage ℓ , let σ_c^ℓ be the standard deviation of its mean response over the reference images $\mathcal{R}$. For a clean test image x and its corrupted version $\tilde{x}$, the shift of the channel’s mean response is

$$\delta_c^\ell = \frac{|\mu_c^\ell(\tilde{x}) - \mu_c^\ell(x)|}{\sigma_c^\ell}, \quad (2)$$

and the shift of its peak-to-mean ratio is defined in the same way, with the standard deviation of ρ . A shift of 1 means that the corruption moved the channel by one standard deviation

of its variation across clean scenes. We average the shift over all channels of a stage and over the test images.

As Fig. 2 shows for the four stages of RT-DETRv2-R18, averaged over the 19 corruption types of [11] at five strengths, the shift of both numbers decreases with depth and is smallest at stage 4. Which early stage moves most, however, depends on the corruption type. Noise, fog and contrast reduction move stage 1 most, whereas elastic transform, pixelation and JPEG compression move stages 2 or 3 most. We therefore measure at stages 1 to 3.

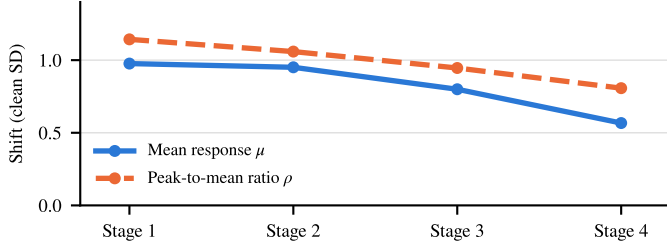


Fig. 2. How far corruptions move the channels of RT-DETRv2-R18 at each stage, in clean standard deviations, for μ and ρ averaged over the 19 corruption types at five strengths.

C. What to Compare With

Sec. III-B measured each shift against the clean version of the same image. At inference, however, the monitor receives only the test image, and no clean version of it exists to compare with. It needs a substitute, built from the clean reference images alone. The simplest one is the mean over all reference images. A test image, however, differs from this mean for two reasons, its scene and its corruption. A busy street, for example, drives edge channels more strongly than an empty sky, even when both images are clean. To isolate the corruption, we compare the test image only with reference images of a similar scene.

This requires a description of the scene that a corruption leaves largely intact. Along the detector, the feature maps relate more and more to its output, the objects in the scene, so we expect the deepest stage to describe the scene best. Table I tests each stage as a key by retrieving, for every test image, the 50 reference images nearest in the stage’s standardized channel means. Shared categories measures whether these neighbours show the same kind of scene, as the overlap (Jaccard index) of their COCO panoptic categories [36] with those of the test image, relative to random reference images. Found again is the share of the neighbours that are retrieved again when the test image is corrupted. The last stage, stage 4, does best on both, so we use the channel means of the last stage, $\ell = L$, as the content key. Even when a corruption changes some of the neighbours, their mean stays clean, since they are always clean images of a similar scene. Under mild fog, for instance, the stage-1 peak-to-mean ratios of the test image fall by 0.83 standard deviations on average, while those of its neighbours’ mean fall by only 0.06.

TABLE I
EACH BACKBONE STAGE OF RT-DETRv2-R18 AS THE CONTENT KEY.

Stage	1	2	3	4
Shared categories	1.7×	1.8×	2.1×	2.6×
Found again	24%	24%	29%	42%

For a test image x , clean or corrupted, let $\mathcal{N}(x) \subset \mathcal{R}$ be its $k = 50$ nearest reference images in this key. Their mean replaces the clean version in Eq. (2), so that the deviation of channel c at stage $\ell \in \{1, 2, 3\}$ is

$$d_c^\ell(x) = \frac{1}{\sigma_c^\ell} \left(\mu_c^\ell(x) - \frac{1}{k} \sum_{r \in \mathcal{N}(x)} \mu_c^\ell(r) \right), \quad (3)$$

and likewise for ρ . Sec. III-D combines these deviations into one score.

D. How to Combine

IV. EXPERIMENT

V. CONCLUSION AND LIMITATION

In summary, we developed a monitor for answering the question “when is the detector not reliable anymore?”.

It is usable for almost all the detector, not constraint within one type of detectors.

It has better performance (some numbers)

However, as we already discussed in sec xx, the model are trained to produce similar activation under xx from the augmentation. So even though their performance might drop, the activation cannot give us much cues, and the monitor cannot work with those yet.

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